CSE430 DSP

Theory Assignment 1

Name: Subhi Bhuiyan

ID: 17201116

Section : 01

Q. Examine on linearity, time variance, causality, and stability of the following signal with appropriate mathematical analysis.

$$y(n) = x(n)\,u(n-10)$$

Linearity

$$x(n) \longrightarrow \text{System} \longrightarrow y(n) = x(n)\,u(n-10)$$

Now,

$$y_1(n) = x_1(n)\,u(n-10) \text{ ——— (i)}$$

$$y_2(n) = x_2(n)\,u(n-10) \text{ ——— (ii)}$$

Addition of two outputs yields ((i) + (ii)),

$$y_1(n) + y_2(n) = x_1(n)\,u(n-10) + x_2(n)\,u(n-10)$$

Now,

$$x_1(n) \rightarrow \text{sys.} \rightarrow y_1(n) = x_1(n)\,u(n-10) \text{ ——— (iii)}$$

$$x_2(n) \rightarrow \text{sys.} \rightarrow y_2(n) = x_2(n)\,u(n-10) \text{ ——— (iv)}$$

Addition of two inputs yields ((iii) + (iv)),

$$x_1(n) + x_2(n) \rightarrow \text{Sys} \rightarrow y'(t) = x_1(n)\,u(n-10) + x_2(n)\,u(n-10)$$

Since the results are same, the system follows the law of additivity.

Scanned with CamScanner

Now, multiplying the output of the system by a constant k,

$$x(n) \rightarrow Sys \rightarrow y(n) \rightarrow 'k' \rightarrow ky(n) = k\,x(n)\,u(n-10)$$

Multiplying the same constant k to the input,

$$k\,x(n) \rightarrow sys \rightarrow y'(n) = k\,x(n)\,u(n-10)$$

Since the results are the same, the system follows the law of homogenity.

Since the system follows both the laws of additivity and homogenity, therefore, it obeys the Principle of Superposition.

So, the system is Linear. (Ans.)

Causality

$$y(n) = \underset{\text{present}}{x(n)}\ \underbrace{u(n-10)}_{\text{past}}$$

Here, the output is dependent on the present and past values of input.

Since the output is independent of future values of input, therefore the equation's nature can be termed as Causal. (Ans.)

Stability

$$x(n) \rightarrow Sys \rightarrow y'(n) = x(n)\,u(n-10)$$

For a bounded input, let's take a dc value of 4.

Now,

$$4 \rightarrow Sys \rightarrow y'(n) = 4.u(n-10) \rightarrow \text{unbounded output}$$

Since for a bounded input, the output of the system is unbounded; therefore, the system is Unstable (Ans.)

Scanned with CamScanner

Time Variance

$$x(n) \rightarrow Sys \rightarrow x(n)\,u(n-10) = y(n)$$

Delaying output by n_0,

$$y(n) \xrightarrow{n_0} y(n-n_0) = x(n-n_0)\,u(n-n_0-10)$$

Delaying input by n_0,

$$x(n) \xrightarrow{n_0} x(n-n_0) \rightarrow Sys \rightarrow x(n-n_0)\,u(n-10)$$

[$u(n-10)$ coefficient is a function, so time variant]

Since the effect of delay is different for the input and output of the system.

So, the system is Time Variant. (Ans)

Scanned with CamScanner